```
import matplotlib.pyplot as plt
```

```
x = 2       # current position
lr = 0.01   # learning rate  (1%)
precision = 0.000001 # precision
step_size = 1 # step size for iteration
max_iter = 10000 # max iteration
iters = 0 # iteration starting from 0
gf = lambda x : (x + 3) ** 2  # gradient function which is given in the ps
```

```
gd = []
while precision < step_size and iters < max_iter:
    prev = x
    x = x - lr * gf(prev)
    step = abs(x - prev)
    iters +=1
    print("Iteration : ", iters, " Value : ", x)
    gd.append(x)
```

```
Iteration :  9943  Value :  -2.9899691424325496
Iteration :  9944  Value :  -2.989970148613585
Iteration :  9945  Value :  -2.9899711545927734
Iteration :  9946  Value :  -2.989972160370175
Iteration :  9947  Value :  -2.9899731659458517
Iteration :  9948  Value :  -2.9899741713198633
Iteration :  9949  Value :  -2.9899751764922704
Iteration :  9950  Value :  -2.989976181463134
Iteration :  9951  Value :  -2.9899771862325144
Iteration :  9952  Value :  -2.9899781908004726
Iteration :  9953  Value :  -2.9899791951670687
Iteration :  9954  Value :  -2.9899801993323636
Iteration :  9955  Value :  -2.9899812032964177
Iteration :  9956  Value :  -2.9899822070592914
Iteration :  9957  Value :  -2.9899832106210456
Iteration :  9958  Value :  -2.98998421398174
Iteration :  9959  Value :  -2.989985217141436
Iteration :  9960  Value :  -2.9899862201001928
Iteration :  9961  Value :  -2.9899872228580717
Iteration :  9962  Value :  -2.9899882254151326
Iteration :  9963  Value :  -2.989989227771436
Iteration :  9964  Value :  -2.989990229927042
Iteration :  9965  Value :  -2.989991231882011
Iteration :  9966  Value :  -2.9899922336364035
Iteration :  9967  Value :  -2.9899932351902794
Iteration :  9968  Value :  -2.989994236543699
Iteration :  9969  Value :  -2.9899952376967227
Iteration :  9970  Value :  -2.98999623864941
Iteration :  9971  Value :  -2.9899972394018217
Iteration :  9972  Value :  -2.9899982399540175
Iteration :  9973  Value :  -2.989999240306058
Iteration :  9974  Value :  -2.990000240458002
Iteration :  9975  Value :  -2.990001240409911
Iteration :  9976  Value :  -2.9900022401618442
Iteration :  9977  Value :  -2.990003239713862
Iteration :  9978  Value :  -2.990004239066024
Iteration :  9979  Value :  -2.990005238218391
Iteration :  9980  Value :  -2.9900062371710217
Iteration :  9981  Value :  -2.9900072359239767
Iteration :  9982  Value :  -2.9900082344773153
Iteration :  9983  Value :  -2.990009232831098
Iteration :  9984  Value :  -2.9900102309853844
Iteration :  9985  Value :  -2.9900112289402343
Iteration :  9986  Value :  -2.990012226695707
Iteration :  9987  Value :  -2.990013224251863
Iteration :  9988  Value :  -2.9900142216087615
Iteration :  9989  Value :  -2.9900152187664624
Iteration :  9990  Value :  -2.990016215725025
Iteration :  9991  Value :  -2.9900172124845095
Iteration :  9992  Value :  -2.990018209044975
Iteration :  9993  Value :  -2.990019205406482
Iteration :  9994  Value :  -2.990020201569089
Iteration :  9995  Value :  -2.990021197532856
Iteration :  9996  Value :  -2.990022193297843
Iteration :  9997  Value :  -2.9900231888641087
Iteration :  9998  Value :  -2.990024184231713
Iteration :  9999  Value :  -2.9900251794007153
Iteration :  10000  Value :  -2.990026174371175
```

```
print("Local Minima : ", x)
```

```
Local Minima :  -2.990026174371175
```

```
plt.plot(gd)
```

```
[<matplotlib.lines.Line2D at 0x79f91a785c40>]
```

